MARC ZGHEIB

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https://marczgheib1.github.io/E-Portfolio/

Summary
Innovation driven product design engineer with 5 years' training in Product Design Engineering and Mechanical Engineering. Skilled in 3D modeling (CAD), prototyping, user centric designing, and innovative design thinking.

EDUCATION

Imperial College London, UK |Dyson School of Design Engineering **Sep 2025 - Sep 2026**
MSc in Design Engineering | with Merit

- Received the prestigious fully funded CMA CGM scholarship, given to only 6 students in 2025.
- Dissertation: Sonic Insoles - wearable gait retraining system, sonified e-textile pressure sensing.
- Key modules: Sensing and Internet of Things (S-IoT), Audio Systems Design, Visual Systems Design, Innovation Management, Contextual Design Engineering, Masters' Thesis Project…
- Competitive League Padel Player with IC Padel Club.

Lebanese American University (LAU) - Byblos, Lebanon **Aug 2021 - Jun 2025**
BEng in Mechanical Engineering | High Distinction | GPA 3.9/4.0

- Granted the excellence fully funded HES scholarship under the USAID Scholarship Program (USP VIII) (2% acceptance rate).
- Consistently placed on the Dean's High Distinction List, completed the University's Honors Program.
- Received the Young Researcher Award – 2025 for publishing paper in the "Energy Conversion and Management" journal, titled "Machine Learning Applications in Optimizing the Performance of Darrieus Wind Turbines".
- Vice President for Astronomy club, robotics club, ASME club.
- Key Modules: Mechatronics Systems Design, CAD, Control Systems, Thermodynamics, HVAC, Renewable Energy, Composite Materials…

PROFESSIONAL EXPERIENCE

UNIPAK Tissue Mill – INDEVCO Group | Mechanical Engineering Trainee **Jun 2024 – Sep 2025**

- Executed 20+ successful repairs in the Maintenance Department and the Powerplant Department.
- Performed preventive maintenance, troubleshooting and repairs of tissue machine mechanisms.
- Completed successful generator maintenance, including engine overhauls of pistons and cylinder heads, by diagnosing mechanical issues and replacing worn parts.
- Conducted vibration testing, load testing and oil/coolant analysis to assess equipment condition.
- Maintained technical documentation and maintenance scheduling and compliance with safety procedures.

Lebanese American University Engineering School | Student Ambassador **May 2023 - Aug 2023**

- Directed and guided prospective students during LAU's Engineering Summer Camp.
- Led campus tours and introduced students to university facilities and student life.
- Coordinated daily activities and supported smooth camp operations.
- Served as a point of contact for students, addressing questions and ensuring their safety.
- Collaborated with faculty and camp staff to deliver a positive student experience.

Association of Energy Engineers Lebanon Chapter | Engineering Trainee May 2022 – Aug 2022

- Collaborated with 5 engineering students on a hydropower project in Damour.
- Performed turbine calculations and simulations for hydropower generation.
- Coordinated team tasks and project milestones to ensure timely delivery.
- Expanded technical knowledge through masterclasses covering energy, sustainability and AI.

KEY DESIGN PROJECTS

Full portfolio of design projects on my website: https://marczgheib1.github.io/E-Portfolio/

- **Motorized Solar-Tracking Patio Umbrella:** Built a 2.5 m steel structure with a motorized sun-tracking mechanism, and solar panels. Focused on integrating dynamic assemblies, solar positioning algorithms and electrical components. Applied SolidWorks, steelmaking, manufacturing, welding, SketchUp, AutoCAD, Matlab.
- **Wearable Heat Therapy Device:** Designed a wrist-worn arthritis relief prototype incorporating custom-designed flexible PCB heating, Arduino Nano BLE control, haptic feedback and LED indicators within a Neoprene strap. Used KiCad, SolidWorks, C++, textile stitching, electronic safety.
- **Assistive Grip & Wrist Trainer:** Invented a handheld rehabilitation device featuring gamified grip/wrist exercises for elderly users and individuals with reduced dexterity, applied SolidWorks, Keyshot, Blender, Matlab.
- **IoT-Enabled Pan-Tilt Laser Turret:** Integrated Arduino C++ and Raspberry Pi, camera-based face tracking and emotion recognition using Python and OpenCV, with servo control, MQTT telemetry, SQL data storage and a live analytics interface. Used SolidWorks, 3d printing, and mechanical junctions.
- **Wearable Gait Retraining System:** Developed a pressure-sensing insole system for gait rehabilitation using plantar pressure measurements and real-time sensory feedback. Mastered SolidWorks, 3d printing, BELA GEM stereo board, BeagleBoard, C++, NumPy, e-textile resistive pressure sensors.
- **Multi-Terrain Emergency E-Scooter:** Created a three-wheeled electric scooter concept for first responders, focused on mobility across varied terrain. Used SolidWorks, Autodesk Fusion 360, Keyshot, Onshape.
- **Real-Time Video-to-Music System:** Built a device with a computer-vision pipeline integrating Raspberry Pi+Arduino hardware with Max/MSP for real-time audio synthesis and visual-to-audio translation. Implemented 3d printed with SolidWorks.

LANGUAGE SKILLS: **Trilingual in English – French – Arabic** | All Native Level

TECHNICAL SKILLS: SolidWorks, MATLAB, LabVIEW, Prototyping, AutoCAD, KiCad, KeyShot, C++, Max/MSP, Python, Arduino, Blender, ANSYS, Homer, Primavera, HAP, Fusion 360, MQTT, Revit, FEA, CFD

CERTIFICATES

SolidWorks accreditation of the Associate Mechanical Design (2022) - Young Researcher Award (2025) - University of Geneva: “Particle Physics: an introduction” (2023) - Georgia Institute of Technology: “Machine Design” (2024) - Top10 in L’Oréal Brandstorm (2024) - Model United Nations (MUN) (2015).

EXTRACURRICULARS

450+ Volunteering Hours – HES Mentorship – Undergraduate Research with Dr. Amne el Cheikh (Thermodynamics & Renewable Energy) – Wrote articles about tech & innovation (HELM Magazine) – Facilitated “Power of space” workshop for LEE program - Padel – Guitar Player – Hiking, Backpacking